Student:	Email:
James Collins	collins222@usf.edu

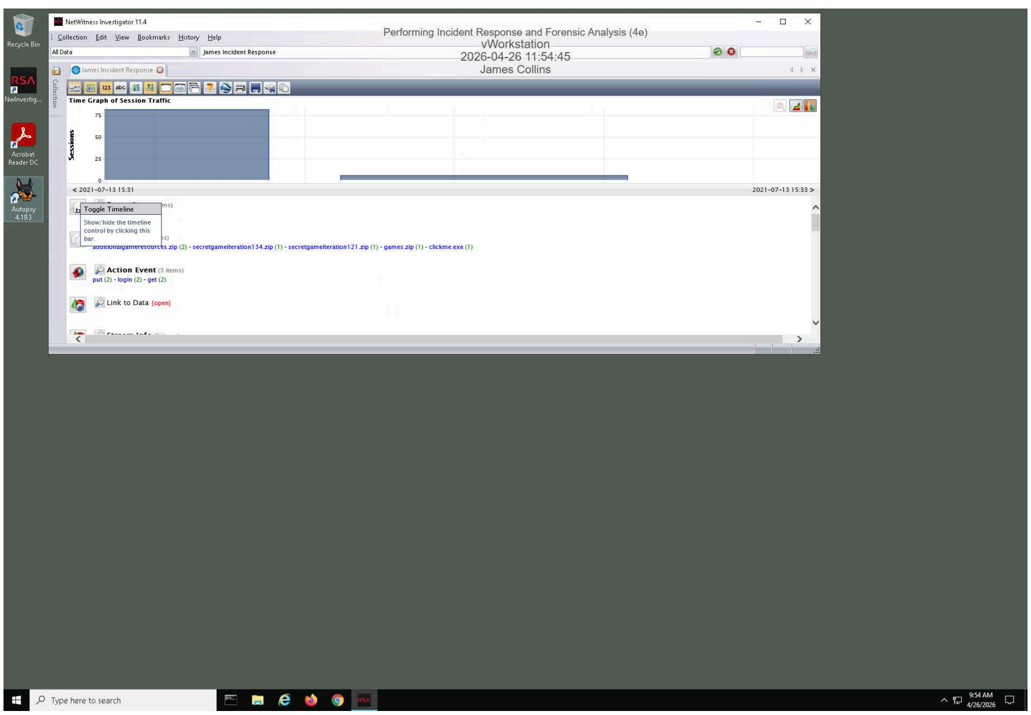
Time on Task:	Progress:
1 hour, 20 minutes	100%

Report Generated: Sunday, April 26, 2026 at 12:58 PM

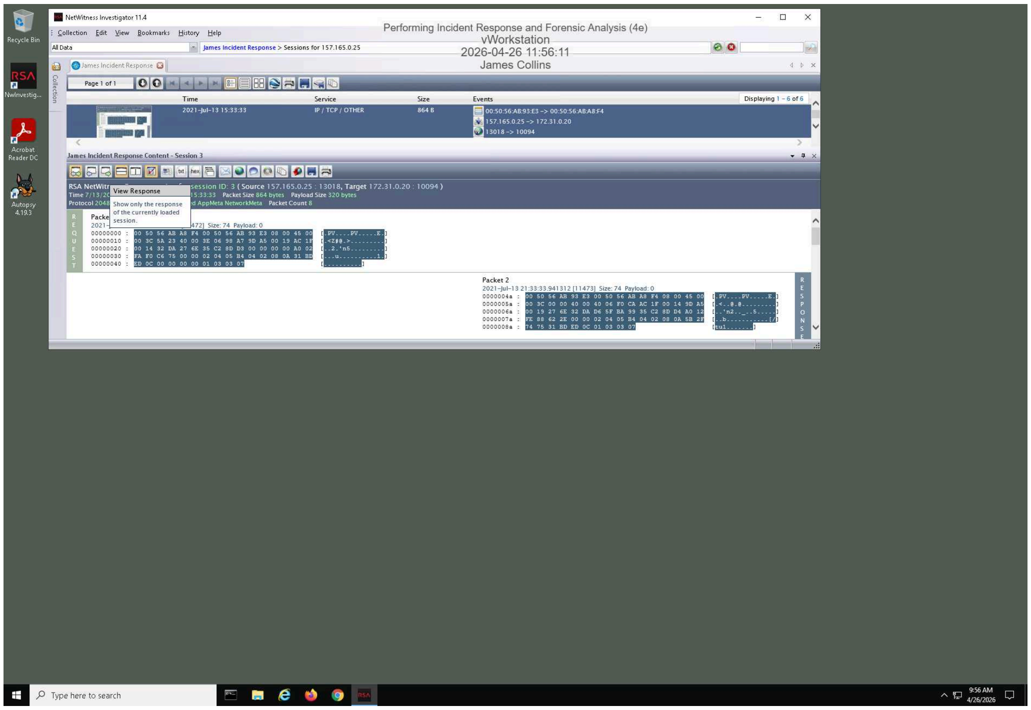
Section 1: Hands-On Demonstration

Part 1: Analyze a PCAP File for Forensic Evidence

10. **Make a screen capture** showing the **Time Graph**.

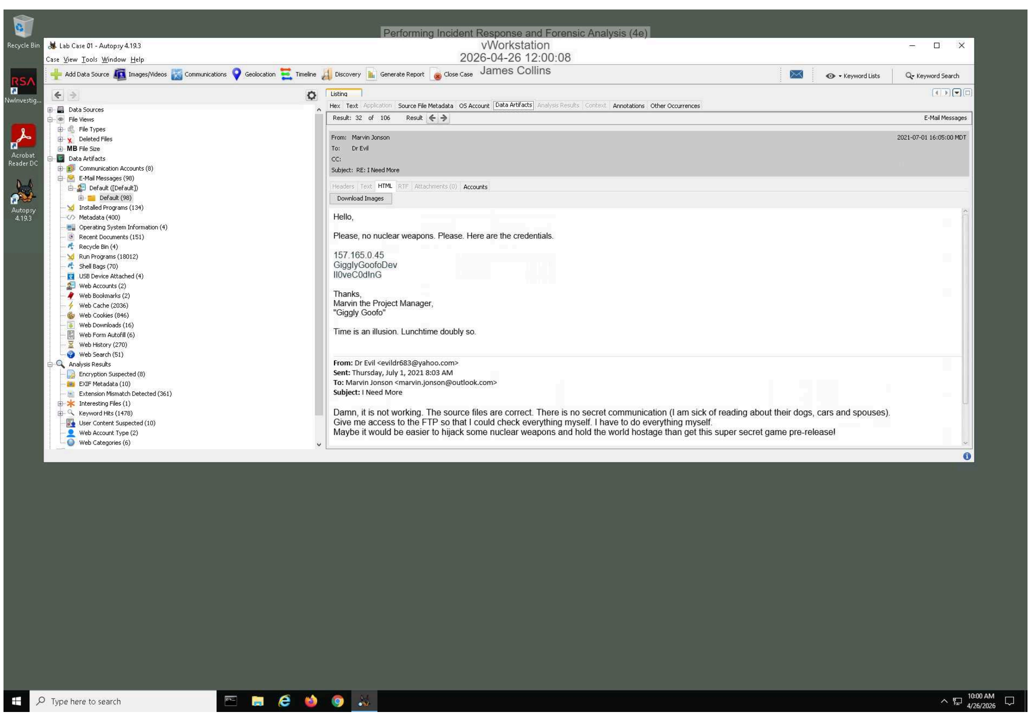


16. **Make a screen capture** showing the **details of the 2021-Jul-13 15:33:00 session**.



Part 2: Analyze a Disk Image for Forensic Evidence

6. **Make a screen capture** showing the **email message containing FTP credentials and the associated timestamps**.



Part 3: Prepare an Incident Response Report

Date

Insert current date here.

April 26,2026

Name

Insert your name here.

James Collins

Incident Priority

Define this incident as High, Medium, Low, or Other.

High

Incident Type

Include all that apply: Compromised System, Compromised User Credentials, Network Attack (e.g., DoS), Malware (e.g. virus, worm, trojan), Reconnaissance (e.g. scanning, sniffing), Lost Equipment/Theft, Physical Break-in, Social Engineering, Law Enforcement Request, Policy Violation, Unknown/Other.

The incident type for this event includes compromised user credentials, social engineering, and policy violation. The incident involves compromised user credentials because Marvin Jonson provided valid FTP login information to an unauthorized external individual. It is classified as social engineering since the attacker used an email to manipulate the user into disclosing sensitive information. Additionally, the incident is considered a policy violation because sharing credentials violates standard organizational security policies and acceptable use guidelines.

Incident Timeline

Define the following: Date and time when the incident was discovered, Date and time when the incident was reported, and Date and time when the incident occurred, as well as any other relevant timeline details.

The incident occurred when Marvin Jonson sent FTP credentials to an external attacker via email, enabling unauthorized access to internal resources. The incident was discovered on 2026-04-26 at 15:56:11 during forensic analysis using NetWitness Investigator, where suspicious FTP traffic and data transfer activity were identified. The incident was reported immediately after discovery once the unauthorized data exfiltration was confirmed. Additional timeline details indicate that following the credential disclosure, the attacker used the provided credentials to initiate an FTP session from the internal system to an external host, resulting in the transfer of data outside the corporate network.

Incident Scope

Define the following: Estimated quantity of systems affected, estimated quantity of users affected, third parties involved or affected, as well as any other relevant scoping information.

The scope of this incident includes a limited number of internal systems and users but poses a significant risk due to data exfiltration. At least one internal workstation, used by Marvin Jonson, and the corporate FTP server were directly affected. The estimated number of users impacted is one confirmed user, though additional users may be indirectly affected depending on the sensitivity of the data stored on the FTP server. A third party, identified as an external attacker (Dr. Evil), was directly involved in receiving the exfiltrated data. The incident extends beyond the internal network due to the transfer of data to an external system, increasing the overall impact and risk to the organization.

Systems Affected by the Incident

Define the following: Attack sources (e.g., IP address, port), attack destinations (e.g., IP address, port), IP addresses of the affected systems, primary functions of the affected systems (e.g., web server, domain controller).

The systems affected by this incident include the internal workstation used by Marvin Jonson with IP address 192.168.1.100, the corporate FTP server with IP address 192.168.1.50 operating over port 21 (FTP), and the external attacker system controlled by Dr. Evil with IP address 174.37.122.17 also communicating over port 21. The attack originated from the internal compromised workstation at 192.168.1.100, which used valid credentials to initiate an FTP connection. The destination of the attack was the external system at 174.37.122.17:21, where the exfiltrated data was transferred. The corporate FTP server at 192.168.1.50:21 was involved as part of the internal infrastructure facilitating the transfer. The method of attack involved credential misuse combined with FTP-based data exfiltration, allowing unauthorized transfer of sensitive data from the internal network to the external attacker system.

Users Affected by the Incident

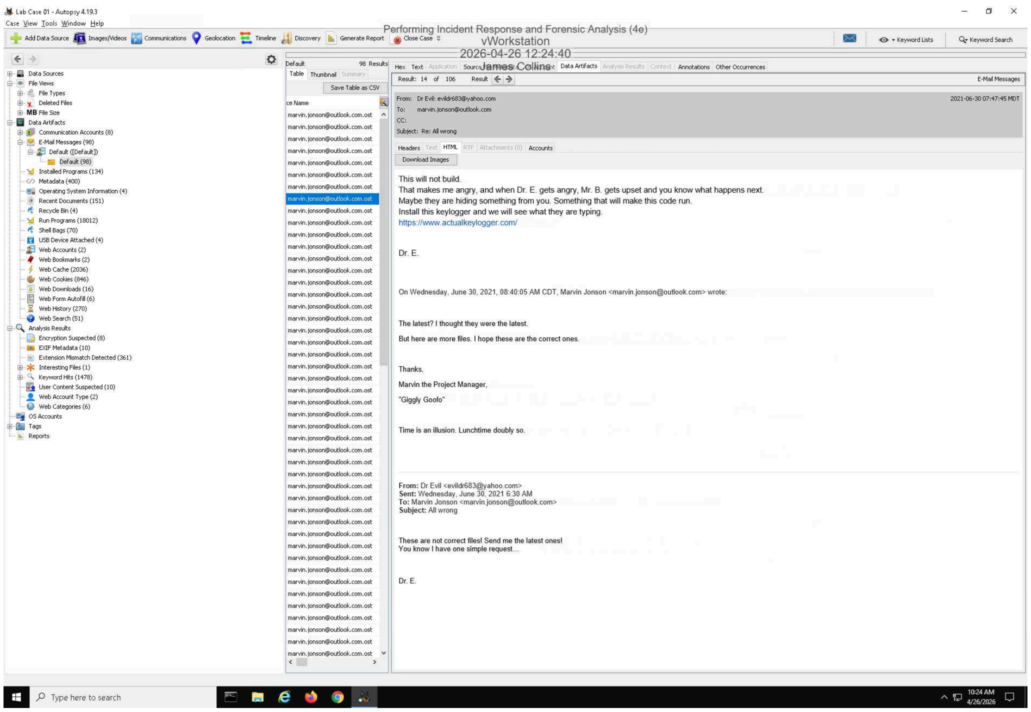
Define the following: Names and job titles of the affected users.

The users affected by this incident include Marvin Jonson, an internal employee whose credentials were compromised and used for unauthorized access and data exfiltration. Marvin Jonson is considered the primary affected user due to his involvement in providing the FTP credentials. Additionally, an external individual identified as Dr. Evil, acting as the attacker, was involved in receiving the exfiltrated data. While no other specific users have been confirmed as directly impacted, additional employees may be indirectly affected depending on the sensitivity and ownership of the data stored on the FTP server.

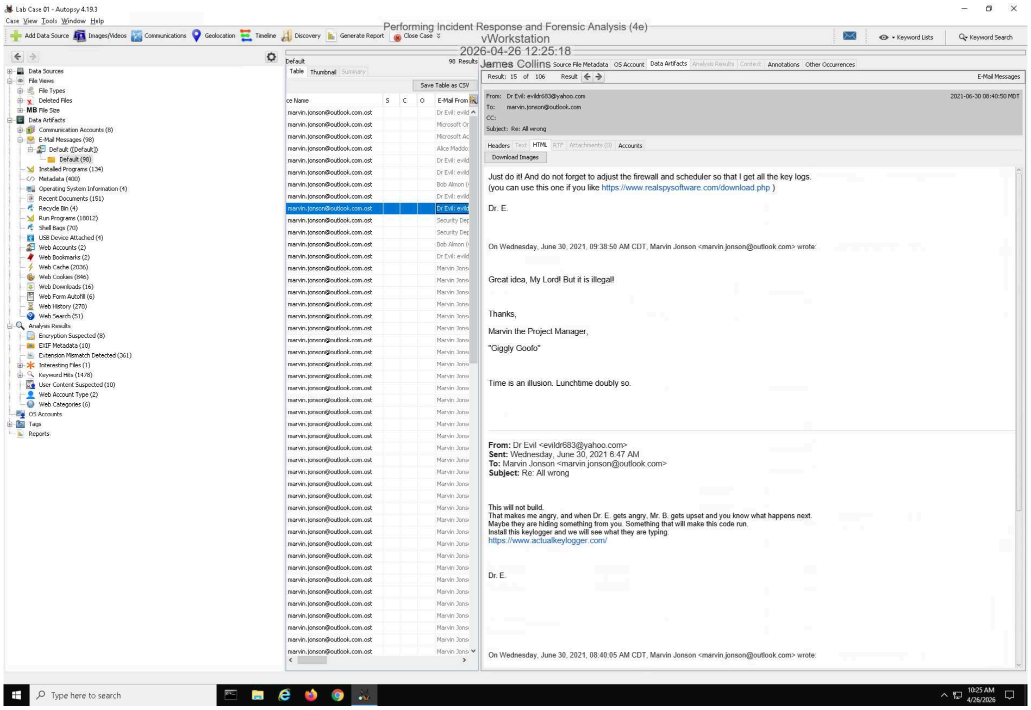
Section 2: Applied Learning

Part 1: Identify Additional Email Evidence

5. **Make a screen capture** showing the **email from Dr. Evil demanding that Marvin install a keylogger**.

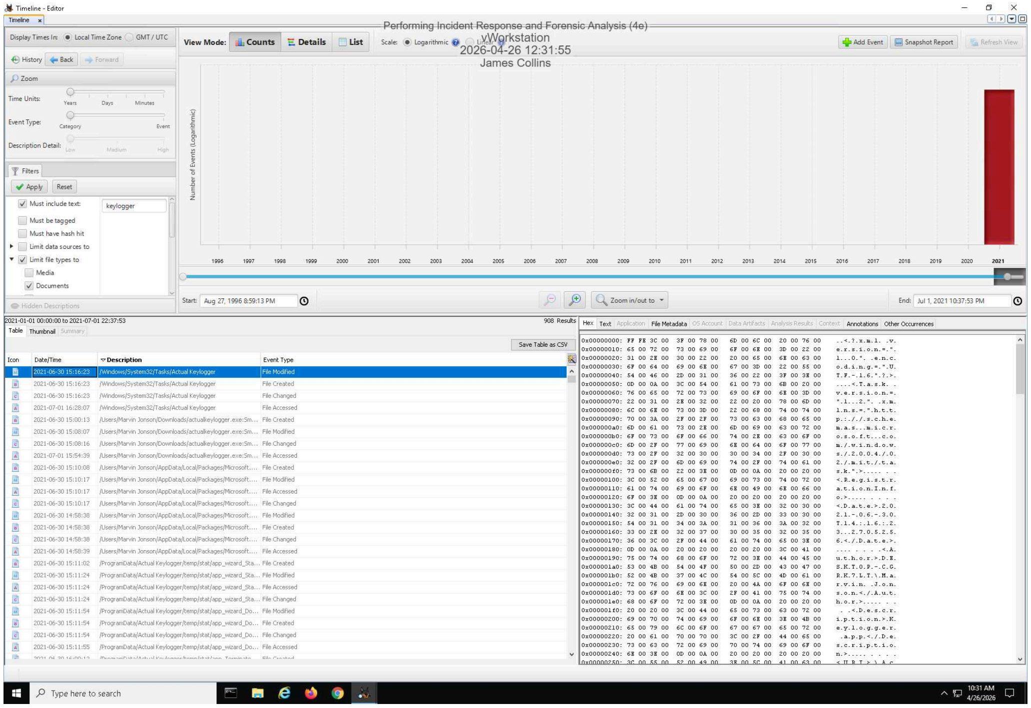


6. **Make a screen capture** showing the **email from Dr. Evil reminding Marvin to update the firewall and scheduler**.

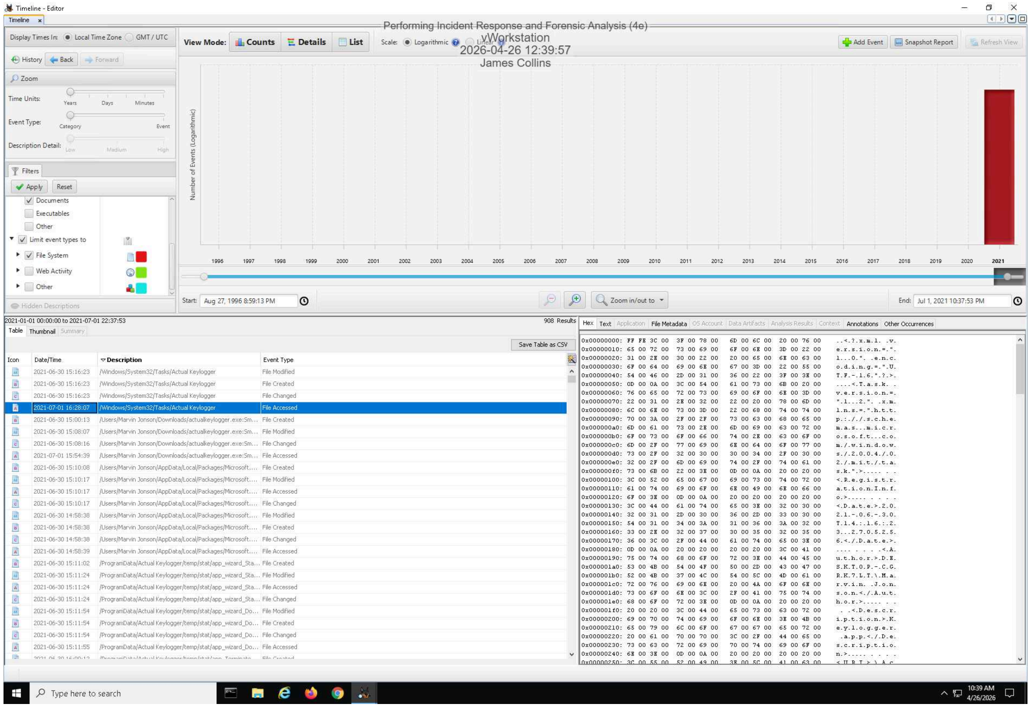


Part 2: Identify Evidence of Spyware

12. **Make a screen capture** showing the **three events that are related to the Actual Keylogger file in the /Windows/System32/Tasks folder with a June 30 timestamp**.



15. **Make a screen capture** showing the **one event that is related to the Actual Keylogger file in the /Windows/System32/Tasks folder with a July 1 timestamp**.



20. **Record** the date and time that the keylogger's executable file was created.

2021-06-30 15:00:13

22. **Record** the date and time when the keylogger's executable file was last started.

2021-07-01 15:54:39

23. **Record** whether you think you have evidence to claim that Marvin opened the keylogger.

There is evidence to support that Marvin Jonson opened the keylogger. The file actualkeylogger.exe located in the Downloads folder shows a File Accessed timestamp of 2021-07-01 15:54:39, which indicates that the file was executed or opened by the user. This access event occurred after the file was created and modified, suggesting user interaction rather than just download activity. Additionally, the metadata confirms the file was downloaded from an external source, further supporting that Marvin interacted with the malicious file, likely leading to credential compromise.

Part 3: Update an Incident Response Report

Date
Insert current date here.

April 26,2026

Name
Insert your name here.

James Collins

Incident Priority
Has the incident priority changed? If so, define the new priority. Otherwise, state that it is unchanged.

The incident priority remains high and is unchanged. While additional evidence has been discovered confirming that Marvin Jonson was socially engineered into installing a keylogger, the overall impact and severity of the incident remain the same due to the continued risk of credential compromise and unauthorized data access.

Incident Type

Has the incident type changed? If so, define any new incident type categories that apply. Otherwise, state that it is unchanged.

The incident type has been updated to include malware in addition to compromised user credentials, social engineering, and policy violation. Newly discovered forensic evidence confirms that Marvin Jonson installed and executed a malicious file identified as actualkeylogger.exe, which indicates the presence of malware on the system. This expands the incident classification beyond credential compromise to include malicious software installation and execution.

Incident Timeline

Has the incident timeline changed? If so, define any new events or revisions in the timeline. Otherwise, state that it is unchanged.

The incident timeline has been updated to include additional events related to the installation and execution of the keylogger. On 2021-06-30 at approximately 15:16:23, the file associated with the keylogger task located in Windows\System32\Tasks\Actual Keylogger was created, modified, and changed, indicating the establishment of persistence. On 2021-07-01 at 15:54:39, the keylogger executable located in the Downloads folder was accessed, suggesting execution by the user. Additionally, on 2021-07-01 at 16:28:07, the scheduled task associated with the keylogger was accessed, further confirming continued activity and persistence on the system.

Incident Scope

Has the incident scope changed? If so, define any new scoping information. Otherwise, state that it is unchanged.

The incident scope has expanded to include additional system-level compromise due to the installation of malware and persistence mechanisms. In addition to the previously identified affected systems, the host system is now confirmed to be compromised through the presence of a keylogger and a scheduled task located in Windows\System32\Tasks\Actual Keylogger. This indicates that the attacker established persistence on the system, increasing the severity of the compromise and potentially allowing continued unauthorized access and monitoring beyond the initial credential theft and data exfiltration.

Systems Affected by the Incident

Has the list of systems affected changed? If so, define any new systems or new information. Otherwise, state that it is unchanged.

The list of systems affected by the incident has been updated to include additional components related to the malware installation and persistence mechanism. In addition to the previously identified internal workstation used by Marvin Jonson, the corporate FTP server, and the external attacker system, the affected systems now include the local host system directories where the keylogger was installed and executed. Specifically, the presence of actualkeylogger.exe in the Downloads directory and the scheduled task located in Windows\System32\Tasks\Actual Keylogger confirm that the host system itself is compromised. This demonstrates that the attack extended beyond credential misuse and data exfiltration to include persistent malware installation on the internal workstation.

Users Affected by the Incident

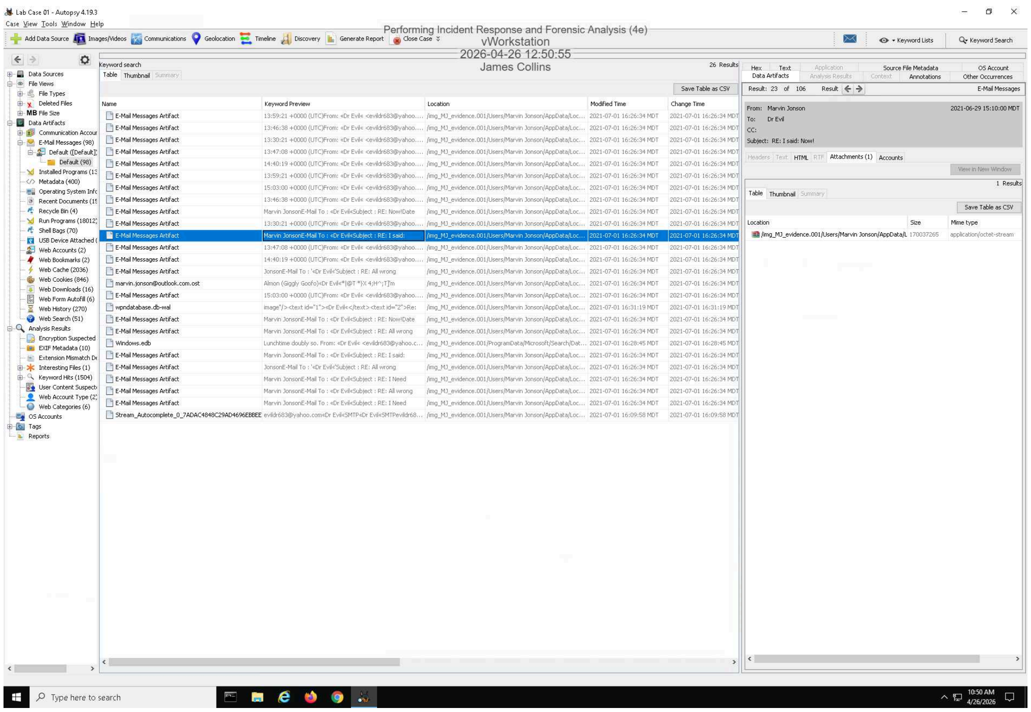
Has the list of users affected changed? If so, define any new users or new information. Otherwise, state that it is unchanged.

The list of users affected by the incident remains unchanged. Marvin Jonson is still the primary affected internal user, as his system was compromised and his credentials were misused. The external attacker, identified as Dr. Evil, remains involved as the unauthorized recipient of the exfiltrated data. No additional users have been identified as directly impacted based on the newly discovered evidence.

Section 3: Challenge and Analysis

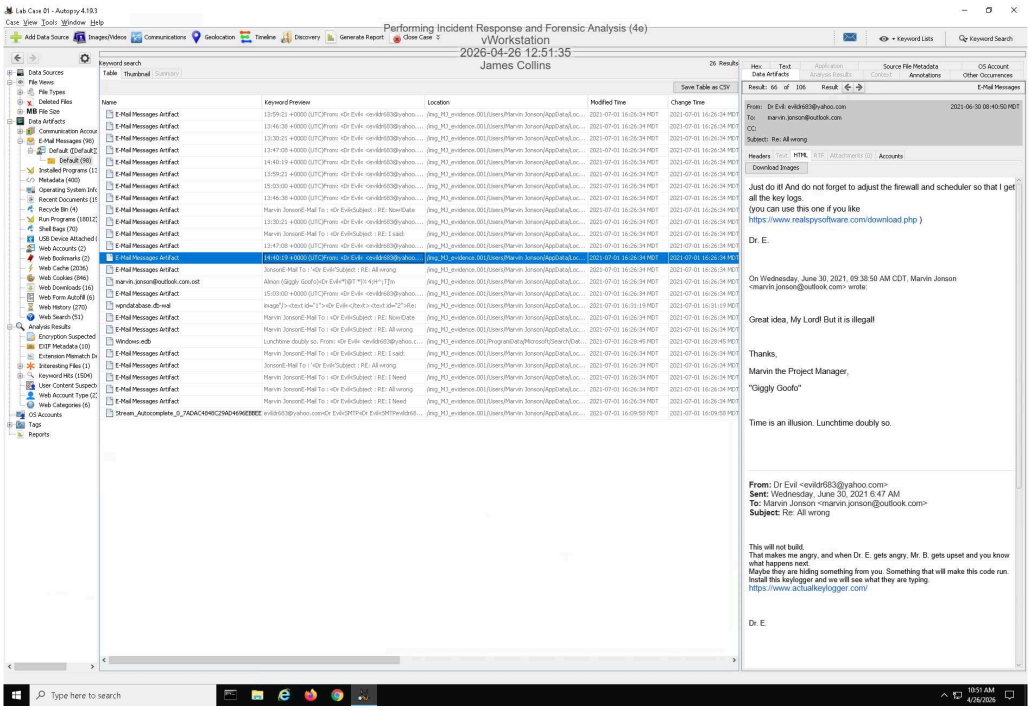
Part 1: Identify Additional Evidence of Data Exfiltration

Make a screen capture showing **an exfiltrated file in Marvin's Outlook database**.



Part 2: Identify Additional Evidence of Spyware

Make a screen capture showing the **email with instructions for installing additional spyware**.



Document the red flags in the email that indicate that it may be a phishing attempt.

The email contains several red flags that indicate it is a phishing attempt. First, the sender claims to be from the “Security Department,” but the email originates from a Gmail address, which is inconsistent with a legitimate corporate security team. Additionally, the message creates a sense of urgency by stating that the software must be installed ASAP and threatens consequences such as fines, which is a common social engineering tactic. The email also includes an external URL that is not associated with the company’s official domain, further indicating malicious intent. Furthermore, legitimate security teams typically deploy updates through automated systems rather than asking users to manually download and install software via email. These inconsistencies strongly suggest that the email is a phishing attempt designed to trick the user into installing malicious software.